

Ritvik Sharma

New York City, NY

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Education

Columbia University

Aug 2025 - Dec 2026

MS in Computer Science - AI & NLP Track

SRM University – Rank: 2/700+

Jun 2019 - May 2023

BTech in Computer Science - AI & ML Track

GPA: 9.86/10

Work Experience

Barclays, UK

Jul 2023 - Jul 2025

Software Engineer-II

Pune, India

- **Saved \$15,000 Monthly:** Innovated utilities ensuring optimal AWS Lambda Usage. **Collaborated** cross-functionally to drive adoption and received Barclays-wide recognition for the achievement. Awarded a **Barclays-Wide Recognition** for the same.
- **Achieved 96% Performance Improvement** in Glue Jobs pertaining to CRUD operations in AWS DynamoDB dealing with 5M+ Records (**12 hours to 27 minutes**). **Led** the solution design using Spark, Dynamic Frames, and Batch Write, **coordinating with platform teams** for seamless implementation.
- **Reduced error-resolution times by 75%** by designing and developing a complex error-routing utility via Python-based Lambdas, IBM Netcool, and SNS filters. Acted as a **key communicator** between development and operations teams to align on system-wide logging strategies.
- **Implemented No-Downtime Deployments:** Innovatively integrated Blue-Green & Canary Based deployments in the existing system. Received **Barclays-Wide Recognition** for the same.
- **Optimized CI-CD Pipelines with 63% higher success rates:** Fostered team collaboration across DevOps and application teams, earning organization-wide recognition at Barclays.
- **Launched Barclays' Major Cloud Platform:** Launched & Led the devising & development of the Integrating layer between Onboarding, Rewards, Payments and other journeys for Barclays' first credit card **ArrowCard** on the Platform.

Projects

VELVÉT - Luxury E-Commerce Website

[Link to Live Demo](#)

- Built a fully functional luxury e-commerce site with React, Gatsby, AWS Lambda, Route53, CloudFront, DynamoDB.
- Synthesized ML-based semantic search by training a Custom BERT Model and hosting on AWS SageMaker. Used AWS OpenSearch for fast product discovery.
- Implemented a secure and high quality AWS backend via IAM Roles, Basic DDoS Prevention and implemented Selenium-based unit + integration tests to ensure Production-Grade Quality.

Inter-University Hackathon Winner: Mercury

[Link to Web/Video Demo](#)

- A Chatbot built utilizing AI/ML & NLP techniques for finding food order details from a sentence such as "The people want **2 Veg Margherita Pizzas** with **Olives** from **Dominos**."
- Designed and Implemented the backend of the Mobile app made with Python, Flutter and using ALBERT (LLM for NLP) and CRF (Probabilistic Prediction) and Speech To Text.

Technical Skills

Languages: Python, C/C++, JavaScript, MySQL

Databases: AWS DynamoDB, MongoDB, SQL Server, Oracle

Developer Tools: Git, GitHub, Jenkins, Docker, Kubernetes, Kafka, Linux

Technologies: AWS(Route53, CloudFront, DataSync, ECS, EC2, SageMaker, OpenSearch, S3, DynamoDB, Lambda), Apache Tomcat, Django, NodeJS, Selenium, gRPC, NLP, BERT, Machine Learning & NLP

Publications & Awards

- Sharma, Ritvik (April 2025). "VectorQuest: Faster and Better Open-Domain Question-Answering System" **Published in Springer Nature Switzerland AG** for the innovation leading to **50 times Faster Prediction**.
- [Inter-University Hackathon Winner](#) for Building an AI-Powered Chatbot for ordering food. [Check Out the Demo Here](#).